Homework 5

Modules 5–6 · Homework · No date

Name _____ Date _____

1. A single 500mg dose of medication was administered orally. The medication is about 88% orally bioavailable and is characterized as having an absorption half-life of approximately 75 minutes, elimination half-life of approximately 4 hours, and apparent volume of distribution of 25 L.

1a Module 5

When does the maximum concentration of drug in the plasma occur?

1b Module 5

What is the expected maximum concentration following this single oral dose?

1c Module 5

What is the expected AUC following this single oral dose?

1d Module 5

If the dose were increased to 1000 mg, when does the maximum concentration of drug in the plasma occur?

1e Module 5

If the dose were increased to 1000 mg, what is the maximum concentration of drug in the plasma that you would expect?

1f Module 5

If the dose were increased to 1000 mg, what is the AUC that you would expect?

2. A 78 kg male patient received multiple IV bolus injections of 15 mg/kg every 8 hours for 48 hours. The medication has an elimination half-life of 4 hours and an apparent

volume of distribution that is 22% of body weight.

2a Module 6

What is the average concentration of drug in the plasma at steady state?

2b Module 6

What is the maximum concentration of drug in the plasma at steady state?

2c Module 6

What is the minimum concentration of drug in the plasma at steady state?

2d Module 6

What is the expected concentration of drug in the plasma 8 hours after administration of the last dose?